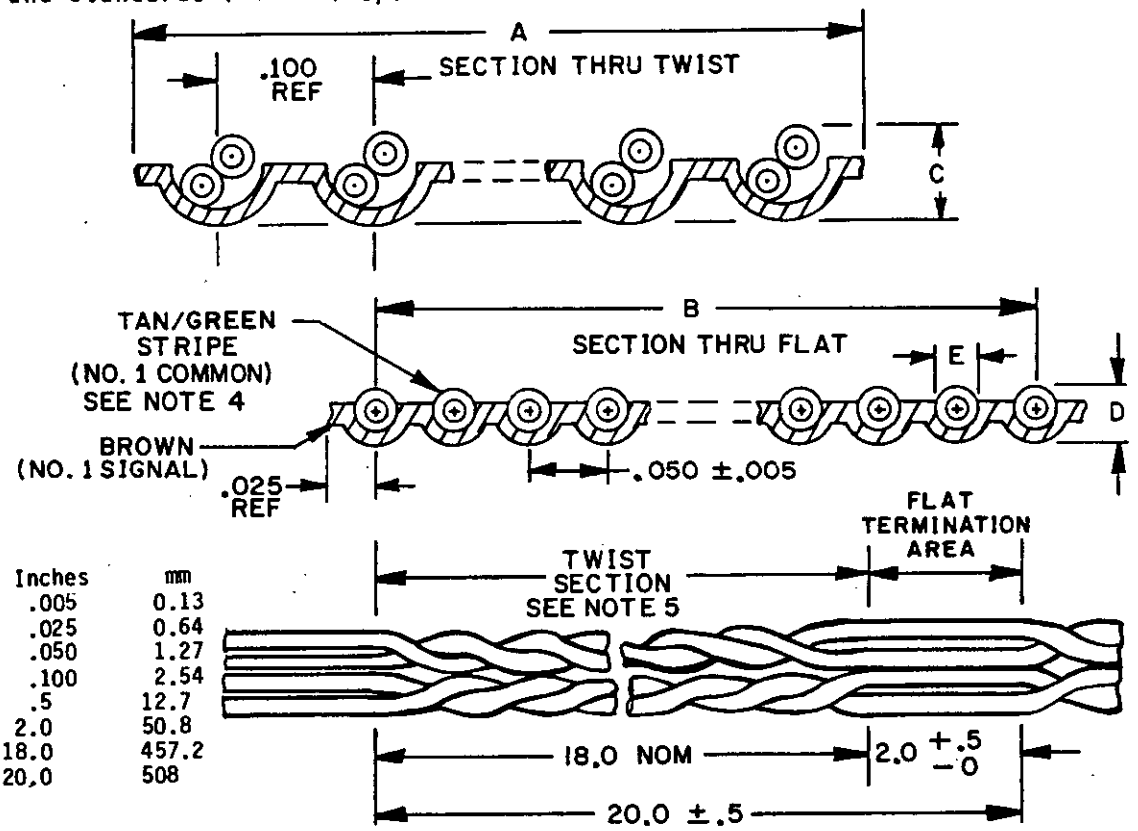
MILITARY SPECIFICATION SHEET

# CABLE, SPECIAL PURPOSE, ELECTRICAL, FLEXIBLE, FLAT, UNSHIELDED, ROUND CONDUCTOR, COLOR CODED, STRANDED, 200°C, MULTIPLE TWISTED PAIRS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-49055.




| Inches | mm |
|---|---|
| .005 | 0.13 |
| .025 | 0.64 |
| .050 | 1.27 |
| .100 | 2.54 |
| .5 | 12.7 |
| 2.0 | 50.8 |
| 18.0 | 457.2 |
| 20.0 | 508 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. All undimensional pictorial representations are for reference purposes only.
4. Striping is required on tan wire next to number 1 wire.
5. The transition area is included in twisted section to assure a full 2 inches (50.8 mm) of .050 (1.27 mm) termination area.

FIGURE 1. Flat cable, twisted pairs with flat termination sections.


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TABLE I. Dash numbers and description. 1/

| Dash number | Number of pairs | AWG | Dimensions | | | | | Weight (lbs/ 1000 ft) | |
|---|---|---|---|---|---|---|---|---|---|
| | | | A | B | C nom | D ±0.003 | E nom O.D. 2/ | Nom | Max |
| 01 | 5 | 28 | 0.500±0.010 | 0.450±0.010 | 0.077 | 0.042 | 0.032 | 18.3 | 19.4 |
| 02 | 7 | 28 | 0.700±0.010 | 0.650±0.010 | 0.077 | 0.042 | 0.032 | 26.9 | 28.5 |
| 03 | 8 | 28 | 0.800±0.015 | 0.750±0.011 | 0.077 | 0.042 | 0.032 | 31.3 | 33.1 |
| 04 | 10 | 28 | 1.000±0.015 | 0.950±0.011 | 0.077 | 0.042 | 0.032 | 39.8 | 42.0 |
| 05 | 13 | 28 | 1.300±0.015 | 1.250±0.011 | 0.077 | 0.042 | 0.032 | 52.8 | 55.7 |
| 06 | 17 | 28 | 1.700±0.020 | 1.650±0.015 | 0.077 | 0.042 | 0.032 | 70.1 | 73.7 |
| 07 | 18 | 28 | 1.800±0.020 | 1.750±0.015 | 0.077 | 0.042 | 0.032 | 74.3 | 78.3 |
| 08 | 20 | 28 | 2.000±0.020 | 1.950±0.015 | 0.077 | 0.042 | 0.032 | 83.0 | 87.2 |
| 09 | 25 | 28 | 2.500±0.020 | 2.450±0.015 | 0.077 | 0.042 | 0.032 | 104.5 | 109.8 |
| 10 | 30 | 28 | 3.000±0.020 | 2.950±0.015 | 0.077 | 0.042 | 0.032 | 126.0 | 132.4 |
| 11 | 32 | 28 | 3.200±0.020 | 3.150±0.015 | 0.077 | 0.042 | 0.032 | 135.2 | 142.1 |
| 12 | 5 | 26 | 0.500±0.010 | 0.450±0.010 | 0.085 | 0.046 | 0.036 | 22.9 | 24.4 |
| 13 | 7 | 26 | 0.700±0.010 | 0.650±0.010 | 0.085 | 0.046 | 0.036 | 34.1 | 36.0 |
| 14 | 8 | 26 | 0.800±0.015 | 0.750±0.011 | 0.085 | 0.046 | 0.036 | 39.6 | 41.8 |
| 15 | 10 | 26 | 1.000±0.015 | 0.950±0.011 | 0.085 | 0.046 | 0.036 | 50.7 | 53.4 |
| 16 | 13 | 26 | 1.300±0.015 | 1.250±0.011 | 0.085 | 0.046 | 0.036 | 67.2 | 70.8 |
| 17 | 17 | 26 | 1.700±0.020 | 1.650±0.015 | 0.085 | 0.046 | 0.036 | 89.5 | 94.2 |
| 18 | 18 | 26 | 1.800±0.020 | 1.750±0.015 | 0.085 | 0.046 | 0.036 | 95.0 | 100.0 |
| 19 | 20 | 26 | 2.000±0.020 | 1.950±0.015 | 0.085 | 0.046 | 0.036 | 106.1 | 111.6 |
| 20 | 25 | 26 | 2.500±0.020 | 2.450±0.015 | 0.085 | 0.046 | 0.036 | 133.9 | 140.8 |
| 21 | 30 | 26 | 3.000±0.020 | 2.950±0.015 | 0.085 | 0.046 | 0.036 | 161.7 | 170.0 |
| 22 | 32 | 26 | 3.200±0.020 | 3.150±0.015 | 0.085 | 0.046 | 0.036 | 172.7 | 183.2 |

1/ Nominal values are for information only, not requirements.
2/ The minimum conductor insulation thickness shall be 0.007 inches.

REQUIREMENTS:

Dimensions and configuration: See figure 1 and table I.

Conductor insulation and laminate material: In accordance with MIL-C-49055 and of the same material.

Voltage rating: 300 volts rms at sea level.

Conductor material: In accordance with MIL-C-49055.

Conductor coating: Silver.

Conductor stranding: Size 26, 7/34 or 28, 7/36 stranding.

Color coding: Ten color repeat (brown, red, orange, yellow, green, blue, violet, gray, white, and black) with common ground as tan with green stripe (see note 4 of figure 1).

Weight: See table I.



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Insulation flaws: 2,500 volts rms.

Dielectric withstanding voltage: 2,000 volts rms.

*Insulation resistance:* *500 megohms* - 1,000 feet (minimum).

Conductor resistance at 23°C:
28 AWG: 69 ohms/1,000 ft. max.
26 AWG: 41 ohms/1,000 ft. max.

Flammability: Nonburning.

Temperature rating: -70°C to 200°C.

Thermal shock: -55°C ±5°C to 200°C ±5°C.

Flexing endurance: -55°C ±5°C; 200°C ±5°C; 300 cycles for 26 AWG, 500 cycles for 28 AWG.

Moisture resistance:

Insulation resistance: 30 megohms/1,000 ft. minimum.

Low pressure (elevated temperature): 200°C ±5°C maximum conductor temperature.

Outgassing: Materials, including inks, shall not exceed a maximum total mass loss of 1 percent and a maximum of 0.10 percent total collected volatile condensable material when tested in accordance with ASTM-E-595.

Laminate or substrate: .005 inch minimum thickness.

High temperature aging: Low temperature 200°C ±5°C; high temperature 230°C ±5°C.

Pair lay length: 1/2 inch nominal lay.

Number of pairs: See table I.

Part or Identifying Number (PIN): M49055/16- (dash number from table I followed by material designator from MIL-C-49055). Example: M49055/16-9H.

## CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - AS
Air Force - 85

Review activities:
Army - AR, AV, MI
Navy - EC, OS, SH
Air Force - 17, 99
DLA - IS, ES
NSA
NAS
NASA - NA

User activities:
Army - AT, ME
Navy - MC

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 6145-1123-3)



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